RentAPlace

Comprehensive Rental Platform

Repository:
`https://github.com/Sravan94-git/RentAPlace`

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Chapter 1

Problem Definition and Objectives

1.1 Problem Statement

Property renting platforms require secure user access, role-based workflows, reliable booking management, and seamless communication between users.

1.2 Objectives

- Build a secure role-based rental platform
- Enable efficient property management
- Provide booking and messaging functionality
- Ensure data integrity and performance

Chapter 2

Frontend & Backend Architecture

2.1 Technology Stack

Frontend:

- Angular
- TypeScript, HTML, CSS

Backend:

- ASP.NET Core Web API
- Entity Framework Core
- JWT Authentication

Database:

- SQL Server

2.2 Architecture

Three-tier architecture:

$$\text{Angular} \rightarrow \text{ASP.NET Core} \rightarrow \text{SQL Server}$$

Chapter 3

Component Breakdown & API Design

3.1 Frontend Components

- Home, Properties, Property Details
- Login, Register
- Booking, Messaging
- Dashboard

3.2 Backend APIs

- AuthController
- PropertyController
- BookingController
- MessageController

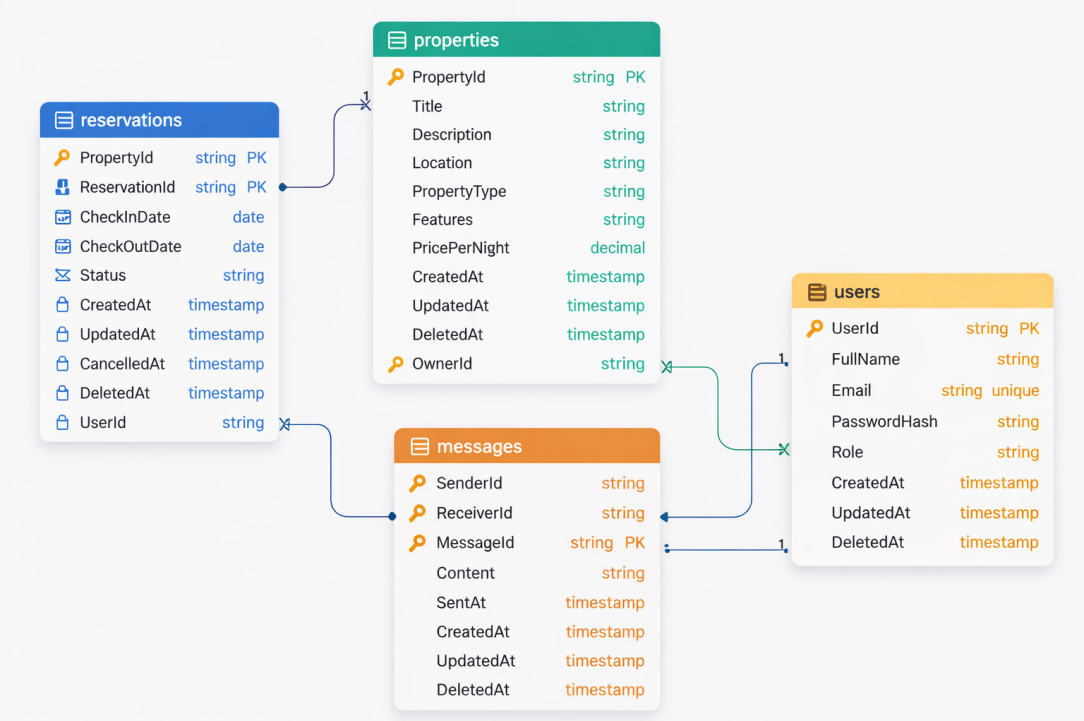
3.3 Authentication

JWT-based authentication with role-based access control.

Chapter 4

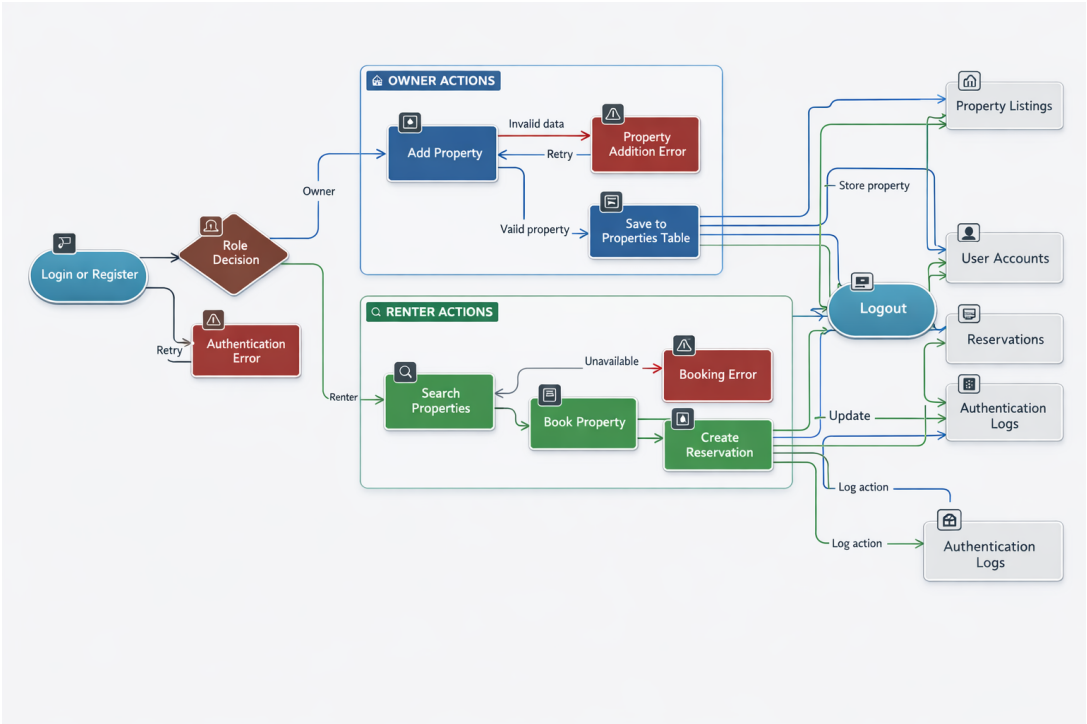
Database Design & ER Diagram

4.1 ER Diagram



Chapter 5

System Flowchart



Chapter 6

Deployment

6.1 Execution

Backend:

```
dotnet run
```

Frontend:

```
npm install
ng serve
```

6.2 Deployment

Deployed using GitHub and VS Code.

Chapter 7

Conclusion and Future Enhancements

RentAPlace successfully demonstrates a modular, scalable full-stack architecture built completely with modern tooling. It executes secure authorization, strict role-based functionality, intuitive UI workflows, interactive communications, and granular analytical insights.

Future Enhancements

- Stripe/PayPal Payment integrations
- Short-lived JWT Refresh token implementations